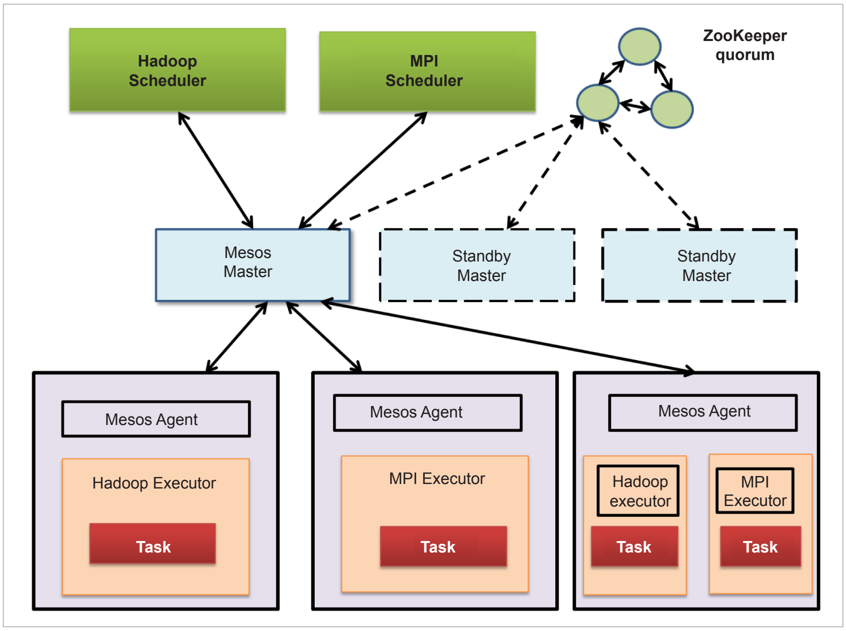

Figure 3: Apache Mesos architecture

parts: scheduler and executor. The scheduler acts as a controller and the executor performs all the activities.

Some of the frameworks provided by Mesos are:

- ***Chronos:*** Fault-tolerant scheduler for Mesos cluster.
- ***Marathon:*** Handles hardware or software failures.
- ***Aurora:*** Service scheduler to run long-running services to take advantage of Mesos' scalability, fault-tolerance and resource isolation.
- ***Hadoop:*** Data processing.
- ***Spark:*** Data processing.
- ***Jenkins:*** Allows dynamic launch of workers on a Mesos cluster depending on the workload.

When to containerise a cloud-native application

Containers, part of a more general software approach called cloud-native, are small software packages which, ideally, perform small, well-defined tasks. Container images include all the software, including settings, libraries and other dependencies, needed for them to run. They are better-suited to frequent changes, for technical or business reasons. This agility aligns them better with cloud architectures.

Cloud-native refers to a set of characteristics and an underlying development methodology for applications and services that are scalable, reliable and have a high level of performance. Containers help accelerate the development and deployment processes, make workloads portable (and even mobile) between different servers and clouds, and are the ideal material from which to build software-defined infrastructure.

Critically, the software infrastructure for containers is primarily open source, written by talented developers and with backing from many large groups.

In some respects, this may seem surprising. Containers don't run only in the cloud. You could run containers on a local, on-premise server if you want—and there are plenty of good reasons to do so.

- You can deploy the same container in any cloud. You can also typically use the same open source tools to manage containers in any cloud. This means that containers maximise mobility between clouds.
- Containers allow you to deploy applications in the cloud without having to worry about the nuances of a particular cloud provider's virtual servers or compute instances.
- Cloud vendors can use containers to build other types of services, such as serverless computing.
- Containers offer security benefits for applications running in the cloud. They add another layer of isolation between your application and the host environment, without requiring you to run a full virtual server.

So, while it's certainly true that you don't need a cloud to use containers, the latter can make cloud-based applications easier to deploy. Containers and the cloud go hand in hand in our cloud-native world. END

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